

The Influence of Spectator Cations on Solvent Reorganization Energy Is a Short-Range Effect

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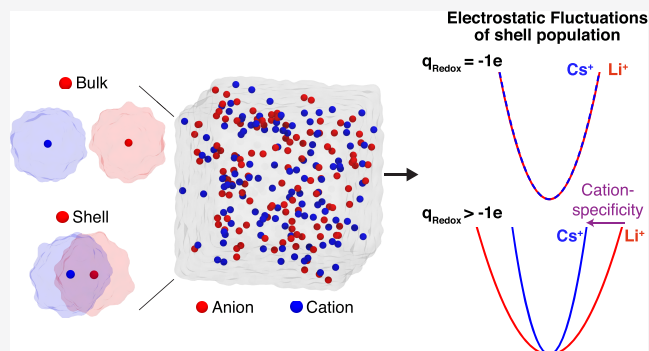
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ABSTRACT: In this manuscript, we use classical molecular dynamics simulation to explore the origin of specific cation effects on the rates of bulk-phase aqueous electron transfer (ET) reactions. We consider 0.6 M solutions of Cl^- and a series of different cations: Li^+ , Na^+ , K^+ , Rb^+ , and Cs^+ . We evaluate the collective electrostatic fluctuations that drive Marcus-like ET and find that they are essentially unaffected by changes in the cationic species. This finding implies that the structure making/breaking properties of various cations do not exert a significant influence on bulk-phase ET reactions. We evaluate the role of ion pairing in these specific cation effects and find, unsurprisingly, that model redox anions that are more highly charged tend to pair more effectively with spectator cations than their monovalent counterparts. We demonstrate that this ion pairing significantly affects local electrostatic fluctuations for the anionic redox species and thus conclude that ion pairing is one of the likely sources of rate-dependent cation effects in aqueous ET reactions.



1. INTRODUCTION

The rate of some aqueous chemical reactions can be systematically varied by changing the identity of the spectator cation. This specific cation effect is often attributed to the differences in the cation's ability to induce additional order or disorder in water's hydrogen bonding network—the so-called “structure making” or “structure breaking” effect.^{1–6} In this manuscript, we evaluate the role of spectator cations in shaping the solvent fluctuations that drive electron transfer (ET) in dilute aqueous electrolyte solutions. Using classical molecular dynamics (MD) simulations, we study a series of alkali chloride solutions and find that the identity of the monovalent cation has essentially no influence on the collective electrostatic fluctuations that determine the solvent contribution to the Marcus ET rate. These results thus imply that specific cation effects in bulk-phase ET reactions likely arise due to systematic variations in the direct interactions of cations with the redox species (e.g., ion pairing and ion bridging effects).

Cation specificity has been observed in many physical properties, including surface interactions,^{7–9} interfacial tension experiments,^{9–12} critical coagulation concentration data of colloidal suspensions,^{13,14} and protein structure and kinetics.^{15,16} One particularly interesting phenomenon is the ability of spectator cations to affect the rates of chemical reactions, which has been observed across a broad range of systems, including both homogeneous^{2,3,17–24} and heterogeneous^{4–6,18–20,22,23,25–31} ET processes. In this work, we narrow our focus to homogeneous ET in the limit of weak coupling, as exemplified by the well-studied self-exchange

reaction between Fe^{2+} and Fe^{3+} complexes.^{32–34} This class of ET reactions can be formalized in the general context of Marcus theory, enabling the physical origins of specific cation effect to be partially isolated and potentially even quantified. In particular, Marcus theory partitions these effects into direct contributions, involving changes to the electronic coupling, and indirect contributions, involving changes in the environmental properties. While the direct contributions are generally system-specific, the indirect contributions are generalizable across different systems and chemical processes (e.g., beyond ET).

Due to the ubiquitous nature of specific cation effect in ET reactions, there have been several experimental and theoretical attempts to determine the mechanistic role of cations in homogeneous ET reactions. Specific cation effects have been observed in ET between anionic species, including octahedral cyanometalate complexes (e.g., $\text{Fe}(\text{CN})_6^{3-/4-}$),^{35–42} tetrahedral redox couples (e.g., $\text{MnO}_4^{2-/3-}$),^{43–45} and square antiprismatic complexes (e.g., $\text{Mo}(\text{CN})_8^{3-/4-}$).² The range of different redox center geometries that exhibit spectator cation sensitivity implies that the mechanism is not reactant-specific

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but rather general. Previous studies have also noted that the homogeneous self-exchange reaction rate is linearly dependent on the concentration of the supporting cations. For instance, Metelski and Swaddle² have used ¹³C NMR measurements to determine the self-exchange rates of Os(CN)₆^{3-/4-}, Mo(CN)₈^{3-/4-}, and W(CN)₈^{3-/4-} couples. They found that in a counterion concentration, [M⁺], range of ~0.001 M to 1 M, the second-order rate constant, *k*_{ex}, is a linear function of the stoichiometric [M⁺] for all the alkali ions with the rate rising in the order of M = Li < Na < K < Rb < Cs. Swaddle et al.⁴² have also studied cation-independent reaction rates by encapsulating the supporting cations with crypt-2.2.2 or 18-crown-6. The cation-dependent reaction path was found to be about 300 times faster than the cation-independent path in 1.0 M K⁺ solutions.

In light of these experimental results, several theories have been put forward to explain the mechanism of cation-induced rate enhancement in homogeneous anion–anion ET.⁴⁶ For instance, it has been proposed that the homogeneous cation catalysis is facilitated by partial desolvation of the supporting cation, which then acts as a tunneling bridge between the two highly charged anions.^{2,23} A three-sphere model was presented to examine the expected effect of cation bridging on the homogeneous rate, but it underpredicts the increase in *k*_{ex} for Fe(CN)₆^{3-/4-}, as the cation is changed from Li⁺ to Cs⁺, and predicts the opposite trend for the dependence of ET rate on R₄N⁺ size as compared to that observed experimentally.²⁰ These observations suggest that the proposed cation bridging mechanism may not be the only source of specific cation effects on ET rates. In this work, we consider alternate mechanisms arising from the influence of cations on the equilibrium fluctuations of the solvation environment.

We simulate a series of 0.6 M bulk aqueous solutions of chloride salts of Li, Na, K, Rb, and Cs. This series of metal cations, listed in order of increasing ionic radius (*i.e.*, decreasing effective charge density), span a range that includes nominal structure-making ions on one end (*e.g.*, Li) and structure-breaking ions on the other (*e.g.*, Cs⁺). Experiments carried out over this same series of electrolyte solutions have shown that the change in ET rate trends with the change in cationic radius.^{2,3,17–20} By studying bulk electrolyte systems in the absence of any specific redox species, we can quantify the effect that different cations have on the collective molecular fluctuations that are known to drive ET reactions.

Our results are presented in two sections. In Section 3.1, we present results generated from bulk electrolyte solutions in the absence of any explicit redox species, focusing specifically on how different cations influence the geometric properties of the aqueous hydrogen bond network and the collective electrostatic fluctuations experienced by molecules within the solution. In Section 3.2, we consider the electrostatic fluctuations experienced by model anionic redox centers and how those fluctuations are influenced by ion pairing effects. These results are preceded by a brief review of Marcus theory and a description of our model and theoretical methods in Section 2.

2. MODEL AND METHODS

2.1. Marcus Theory and the Vertical Energy Gap.

Marcus theory provides a general framework for understanding how solvent fluctuations influence the rate of electron transfer between two weakly coupled molecular species. Specifically, the Marcus ET rate is given by

$$k_{\text{ET}} = \frac{2\pi}{\hbar} |H_{\text{AB}}|^2 \frac{1}{\sqrt{4\pi k_{\text{B}} T \lambda}} e^{-\beta \Delta A^\ddagger} \quad (1)$$

where *H*_{AB} is the electronic coupling (assumed to be weak), λ is the reorganization energy, $\hbar$ is the reduced Planck constant, and $\beta^{-1} = k_{\text{B}}T$ (*i.e.*, the Boltzmann constant times temperature). The activation free energy, $\Delta A^\ddagger$, is determined by solvent fluctuations and can be formulated by considering the intersection of two diabatic free energy surfaces, corresponding to the states where the transferred electron is on one molecule (state A) or the other (state B). These states are resolved along a solvent reaction coordinate, as illustrated in Figure 1. With a suitable model for generating these free energy surfaces, it is possible to isolate and study the solvent's contribution to ET rates.^{47–52}

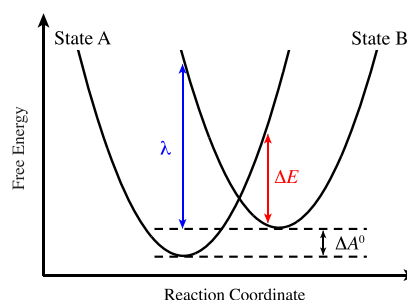


Figure 1. Marcus diabatic energy surfaces for redox electron transfer reactions. In the figure, λ is the reorganization energy, Δ*E* is the diabatic energy difference between the initial and final states in the same molecular configuration, and Δ*A*⁰ is the free energy difference between the reactant and product states.

Remarkable simplification emerges when the solvent coordinate is chosen to be the vertical energy gap, Δ*E* = *E*_B({*r*^N}) – *E*_A({*r*^N}), which defines the potential energy to change diabatic states (*i.e.*, an electron transfer) at fixed nuclear configuration, {*r*^N}. With this choice and an assumption that fluctuations in Δ*E* obey Gaussian statistics, the activation free energy can be expressed as

$$\Delta A^\ddagger = \frac{(\lambda + \Delta A^0)^2}{4\lambda} \quad (2)$$

where Δ*A*⁰ is the reaction free energy and λ is the reorganization energy, as illustrated in Figure 1. The value of λ is related to the width of the parabolic free energy surfaces as

$$\lambda = \frac{\beta \sigma^2[\Delta E_{\text{redox}}]}{2} = \frac{\langle \Delta E_{\text{B}} \rangle - \langle \Delta E_{\text{A}} \rangle}{2} \quad (3)$$

where σ²[Δ*E*_{redox}] is the variance of equilibrium distribution for Δ*E*, and ⟨Δ*E*_A⟩ and ⟨Δ*E*_B⟩ are the average values of Δ*E* for state A and state B, respectively.^{48,49}

In the context of an MD simulation, Δ*E* can be computed by calculating the energy to perform an ET for a given nuclear configuration (without actually performing the ET) and λ can be computed based on the equilibrium statistics of Δ*E* obtained in either diabatic state. In point charge models, where the force field parameters of the oxidized and reduced species differ only in their charges, the vertical energy gap to transfer an electron in diabatic A, *i.e.*, from atom α of a donor molecule to atom β of an acceptor molecule, is

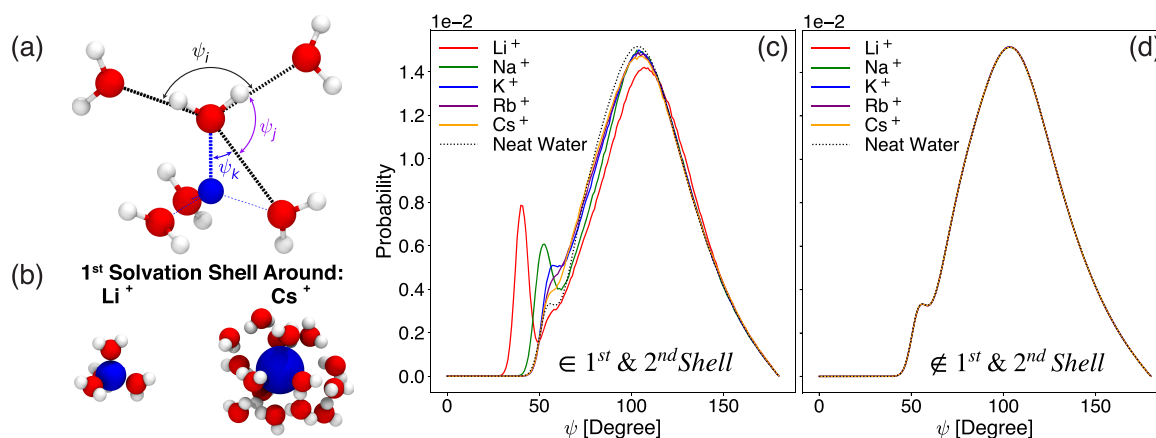


Figure 2. (a) Schematic illustrating three (of six) interbond angles around a water molecule in the first solvation shell of a cation. Water molecules and cations are depicted in red/white and blue, respectively. For this analysis, cations are also counted as hydrogen bonding sites. (b) Representative snapshots of the first solvation shell around Li^+ and Cs^+ , highlighting the difference between the compact, tetrahedral solvation around Li^+ and the more diffuse solvation around Cs^+ . These differences are quantified in plots (c) and (d), which show interbond angle distributions for the water molecules inside and outside the first two solvation shells of the cations, respectively.

$$\Delta E_A = \left[\phi_\alpha - \phi_\beta - \frac{1}{r_{\alpha\beta}} \right] \quad (4)$$

and likewise in diabat B (i.e., ET from β to α)

$$\Delta E_B = \left[\phi_\alpha - \phi_\beta + \frac{1}{r_{\alpha\beta}} \right] \quad (5)$$

where ϕ_i denotes the Madelung potential on atom i and $r_{\alpha\beta}$ is the distance between atoms α and β . The Madelung potential for atom i is defined as

$$\phi_i = \sum_{j \neq i}^N \frac{q_j}{r_{ij}} \quad (6)$$

where the summation is taken over all other atoms in the system.

Based on the equations above, the solvent contribution to the ET rate can be evaluated by sampling the statistics of ϕ_α and ϕ_β . The dependence of these rates on the changes in environmental conditions, such as the identity of spectator ions, is therefore encoded in how these statistics vary.

2.2. Simulation Setup. The simulated bulk systems consisted of 10,738 water molecules, 119 alkali ions (Li^+ , Na^+ , K^+ , Rb^+ , or Cs^+), and 119 Cl^- counterions in a periodically replicated cubic cell. These conditions correspond to a cation concentration of approximately 0.6 M (~ 90 water molecules per ion). Simulation cell size was determined by equilibration under NPT conditions with $T = 300$ K and $P = 1$ atm. Molecular dynamics (MD) simulations were performed in LAMMPS using a 2 fs time step.⁵³ A particle-mesh Ewald algorithm with a real-space cutoff value of 9 Å was used to calculate long-range electrostatic interactions. The temperature and pressure during the equilibration periods were maintained by a Nosé–Hoover thermostat and barostat, respectively. Water molecules were modeled with the SPC/E force field,⁵⁴ and bond constraints were enforced via the SHAKE algorithm.⁵⁵ For ions, the Lennard-Jones parameters from ref 56 were adopted along with a scaled-ionic-charge model to account for electronic polarization effects in a classical nonpolarizable force field.^{57–59} All the parameters used are summarized in Table S1 (see the Supporting Information).

Lorentz–Berthelot mixing rules were used to derive mixed Lennard-Jones parameters. After a density equilibration of 0.1 ns at NPT, cell dimensions were fixed and the system was equilibrated for an additional 0.1 ns at NVT with $T = 300$ K. A final production run of 1 ns was then performed in the NVE ensemble in a cubic simulation box with a side length of approximately 69.2 Å.

In Section 3.2, we investigate the role of redox species' charge in ion pairing. For this investigation, we vary the charge on Cl^- from -1 to -1.5 (corresponding to $-0.94e$ and $-1.41e$, respectively, in polarization-corrected charge values) while keeping the same force fields and equilibration protocols as above for all other atoms. The low-valence system (with -1 charge on the model redox center) consist of 10,738 water molecules along with 119 cations and 119 anions. The high-valence system (with -1.5 charge on the model redox center) consists of the same number of water molecules; however, to maintain charge neutrality, the numbers of ions are changed to 120 and 80 for cations and anions, respectively. Additional simulations using -2 anions were performed to reinforce the conclusion derived from -1.5 anion simulations (see results in the Supporting Information).

Furthermore, to eliminate the influence of ion–ion correlations (see Sections 3.1.1 and 3.1.3), we performed single-cation simulations that consisted of 1 cation and 499 water molecules. The force fields for these simulations were identical to those described above. The system was equilibrated for 0.1 ns at $T = 300$ K and $P = 1$ atm in NPT ensemble, and then the final production run of 1 ns was performed under the same conditions. The side length of the simulation box for the NPT production runs was approximately 24.67 Å. A uniform background charge was used to maintain the charge neutrality of the simulation box.

3. RESULTS AND DISCUSSION

3.1. Specific Cation Effects in Bulk Electrolyte Solutions. Spectator cations can influence ET rates indirectly, by modifying the electrostatic fluctuations of the environment, or directly, through effects associated with ion pairing and complexation. In this section, we address indirect effects focusing specifically on bulk electrolyte solutions (free of any redox species). We first analyze the statistics of the geometric

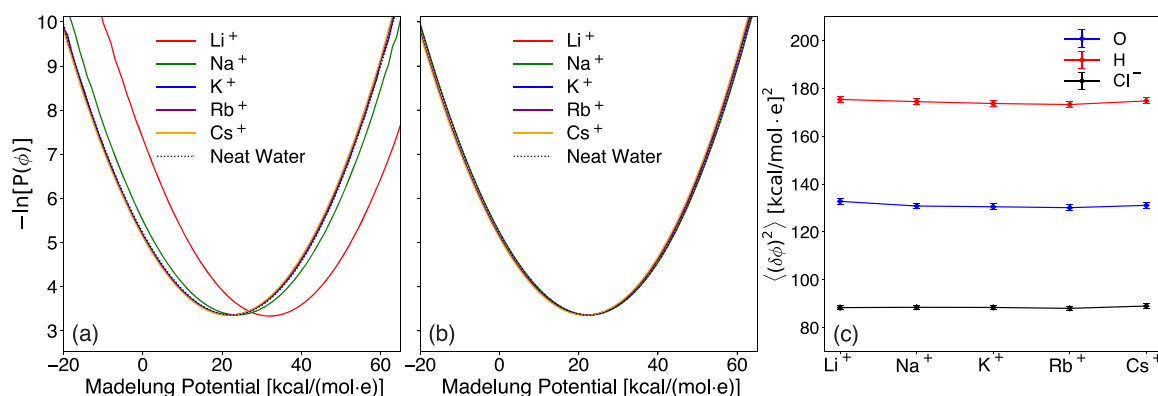


Figure 3. Madelung potential distributions for water molecules: (a) inside and (b) outside the first solvation shell of the cations. Distributions from the simulation of neat water (dotted line) are included as the reference system. Panels (a) and (b) share the same y-axis. (c) Equilibrium fluctuations of the Madelung potential, $\langle(\delta\phi)^2\rangle = \langle(\phi - \langle\phi\rangle)^2\rangle$, of different types of atoms (O, H, and Cl⁻) in the simulation box with different alkali cations (shown as the x-axis).

parameters that characterize the aqueous hydrogen bonding network (Section 3.1.1). Then, we evaluate the fluctuations in the Madelung potentials (see eq 6) ϕ_{Cl^-} for chloride ions (Section 3.1.2) and ϕ_{O} for water molecules (Sections 3.1.2 and 3.1.3). These results combine to indicate a negligible effect of cation identity on the electrostatic fluctuations that are intrinsic to the bulk electrolyte.

3.1.1. Geometric Fluctuations in the Hydrogen Bonding Network Near and Far from Cations. The tendency of various cations to act as structure makers or structure breakers in water is most naturally quantified by analyzing the statistics of hydrogen bonding geometries. Here, in single-cation aqueous systems, we focus specifically on the distribution of the six interbond angles (each denoted as ψ_i) formed by a water molecule and its four nearest neighbors, as depicted in Figure 2a. For clarity, the water molecules in the simulation box are separated into two subpopulations—the waters inside and outside the first two solvation shells of cations.

Example configurations in Figure 2b demonstrate that smaller cations (like Li⁺) tend to order and align the first solvation shell whereas larger cations (like Cs⁺) tend to have disordered and diffuse solvation shells. As a result, the interbond angle distributions of the water molecules in the first few solvation shells of monovalent cations vary systematically with cation size. As Figure 2c illustrates, the deviation of the angle distribution (solid color lines) from that of the bulk (black dotted lines) increases with decreasing cation radius. However, a cation's influence on the geometry of the H-bond network is short ranged, extending to only the second solvation shell. Figure 2d shows that there is no difference among the interbond angle distributions of water molecules beyond the second solvation shell in different alkali solutions. A similar conclusion arises from studying the so-called tetrahedral order parameter, Q_6 ,^{60–63} which is a complementary measure of a hydrogen bond network structure (see Section S3 in the Supporting Information for the statistics of Q_6).

Both interbond angle and tetrahedral order parameter analyses indicate that the change in geometry of an aqueous hydrogen bond network is confined only to the first few solvation shells around the cations. In single-cation systems (used to obtain data in Figure 2), molecules in the first and second solvation shell make up approximately 6% of the overall population of water molecules, while in bulk-ion systems with

~0.6 M alkali halide salt (for data in Sections 3.1.2 and 3.2), this number increases to about 30%. Clearly, cations only make or break structures within their immediate vicinity, leaving the properties of the bulk water region unaffected.

This short-range ion effect is in agreement with previous literature studies in which different theoretical^{64–69} and experimental^{70,71} methods were used to study the influence of ions on surrounding water molecules. In the following subsection, we demonstrate that the electrostatic fluctuations of the bulk water molecules are similarly unaffected by changing cation identity.

3.1.2. Madelung Potential Fluctuations Near and Far from Cations. The statistics of Madelung potential fluctuations determine the Marcus reorganization energy, λ , which in turn determines ET rates (see Section 2.1). The extent to which these fluctuations are modified when the identity of the spectator cation is changed can be identified by characterizing the probability distribution, $P(\phi)$, for a given redox species. Here, we focus on the Madelung potential fluctuations for the oxygen atoms of water molecules in larger simulated systems containing 0.6 M alkali chloride solution. Specifically, we compute $P(\phi_{\text{O}})$ for the subpopulation of water molecules that reside inside or outside of the first cation solvation shell. We define the first solvation shell as the set of water molecules with oxygen positions within a distance cutoff of the cation center governed by the first minimum in the cation–oxygen radial distribution function (see Figure S2), i.e., 2.7, 3.2, 3.5, 3.8, and 4.0 Å around Li⁺, Na⁺, K⁺, Rb⁺, and Cs⁺, respectively. These distributions are presented in Figure 3a,b for inside and outside the shell, respectively.

The results in Figure 3a,b reveal that the influence of different cations on the electrostatic fluctuations of the solvent is isolated to the cationic solvation shell. Water molecules within the first solvation shell of a cation experience a shift in the mean value of $P(\phi_{\text{O}})$ with changes in the cation identity. Considerable shifts are associated with the smaller cations (such as Li⁺), whereas the bigger cations (e.g., Cs⁺) yield almost no shift relative to the bulk. Importantly, even when the mean position of $P(\phi_{\text{O}})$ shifts, its variance, $\langle(\delta\phi_{\text{O}})^2\rangle$ (reflected by the plotted parabola width), remains essentially unchanged. This insensitivity of the variance of ϕ to cation identity is further illustrated in Figure 3c for three different types of particles, namely, O, H, and Cl⁻. In spite of changes in local hydrogen bond geometry, the electrostatic fluctuations felt by

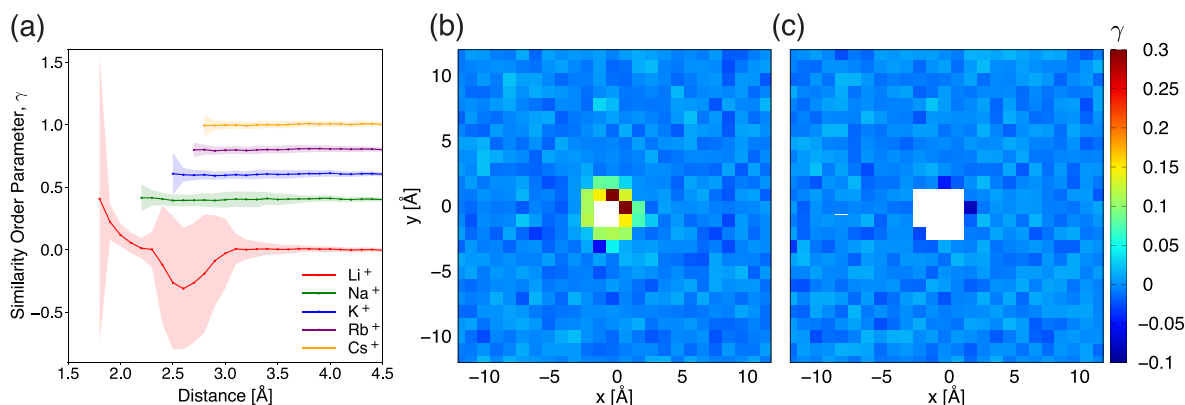


Figure 4. (a) Radially averaged values of γ centered around each cation. The curves for Na^+ , K^+ , Rb^+ , and Cs^+ are vertically shifted by +0.5, +0.7, +0.9, and +1.1, respectively, for clarity. The shaded regions indicate the standard deviations calculated across 50,000 configurations collected over a 1 ns trajectory of single-cation simulation. Spatial maps of γ for Li^+ and Cs^+ are presented in (b) and (c), respectively (see text for details regarding the spatial mapping protocol). The statistics for Na^+ , K^+ , and Rb^+ -containing systems essentially look the same as that for Cs^+ -containing system and are plotted in the Supporting Information (Section 2.2). For both 1D and 2D plots, poorly sampled regions have been omitted, as represented by white pixels in panels (b) and (c). Panels (b) and (c) share the same y-axis and color bar.

species in solution are not strongly affected. Outside of the first solvation shell of the spectator cations, the distribution $P(\phi_{\text{O}})$ is indistinguishable for different cation species, as illustrated in Figure 3b.

The results in Figure 3 demonstrate that despite the geometric influence that cations exert on the hydrogen bonding network, the intrinsic electrostatic fluctuations of a dilute electrostatic solution are not affected by changing the identity of the cationic species. The variance in potential distributions, which determines $\Delta A^\ddagger$ in eq 1, is essentially independent of the identity of the cationic species, even for species in direct contact with a cation. This observation implies that cation-dependent rates in homogeneous ET reactions are likely not the result of changes to the hydration properties of the electrolyte solution. Taken alongside Figure 2, these results refute the notion that cation effects emerge due to the tendency of different cations to make or break structures in the extended aqueous hydrogen bonding environment.

3.1.3. Spatially Resolving a Cation's Influence on Local Potential Fluctuations. To better understand how alkali cations perturb the electrostatic fluctuations of their local aqueous environment, we have performed a comparative statistical analysis of ϕ_{O} fluctuations in the vicinity of cations. In this analysis, we quantify the influence of a cation on its surroundings by computing the likelihood that nearby water molecules would experience the same values of ϕ_{O} in the bulk (*i.e.*, in the absence of any ions). Specifically, we histogrammed the probability distribution of ϕ_{O} from an equilibrium simulation of neat water, $P(\phi_{\text{O}}|\text{bulk})$, and then use this distribution to define the log-likelihood parameter

$$\Gamma(\{\phi_{\text{O}}\}) = -\frac{1}{N_{\text{samp}}} \sum_{i=1}^{N_{\text{samp}}} \ln[P(\phi_{\text{O},i}|\text{bulk})] \quad (7)$$

where $\{\phi_{\text{O}}\}$ denotes a series of values of ϕ_{O} sampled from a single water molecule (*i.e.*, $\phi_{\text{O},1}$, $\phi_{\text{O},2}$, ..., $\phi_{\text{O},N_{\text{samp}}}$) and $P(\phi_{\text{O},i}|\text{bulk})$ is the probability of finding a water molecule with a potential of $\phi_{\text{O},i}$ in the neat water system. We then define a similarity order parameter

$$\gamma(\{\phi_{\text{O}}\}) = \left(\frac{\Gamma(\{\phi_{\text{O}}\})}{\langle \Gamma \rangle_{\text{bulk}}} - 1 \right) \quad (8)$$

where $\langle \Gamma \rangle_{\text{bulk}}$ is the average value of Γ for water molecules in neat water sampled over the same number of configurations (*i.e.*, N_{samp}). Water molecules that experience fluctuations like those of neat water are characterized with values of $\gamma \approx 0$, and molecules that experience nonbulk-like fluctuations yield values of $|\gamma| > 0$, with larger values of $|\gamma|$ corresponding to larger deviations from bulk-like behavior. When applied to water molecules in an electrolyte solution, γ can be used to identify the specific influence of ions on the electrostatic fluctuations of nearby water molecules.

We calculate the values of γ for water molecules in a simulation of a single cation in liquid water (see Section 2.2 for the description of the system) by sampling 50,000 configurations evenly spaced over a 1 ns interval. We then generate a radial function of γ , averaged over values sampled along the radial distance from the center of the cation, as presented in Figure 4a. Although the cations introduce significant geometric perturbation to the aqueous hydrogen bonding network in their solvation shells, they have a negligible influence on the electrostatic fluctuations of water, even inside their first solvation shells. The only exception is the lithium-containing system, as illustrated in Figure 4a (red line), which displays a significant deviation from the reference system in the first solvation shell. This deviation is a trivial consequence of the very rigid nature of the first solvation shell of lithium (*e.g.*, compare the width of first peak in RDF (radial distribution function) for Li^+ and those of the other alkali ions in Figure S2 in the Supporting Information).

For more intuitive data interpretation, we also generated the spatial map of γ by sampling $N_{\text{samp}} = 50,000$ configurations evenly spaced along a 1 ns trajectory. We then averaged over values sampled along the xy -plane through the center of the cation as trajectory by projecting values onto a cubic lattice with the cation at the origin. The data in each grid point represents the average of water molecules sampled from a sphere of radius 1 Å centered at that grid position. The results are presented in Figure 4b,c for Li^+ and Cs^+ , respectively. These plots show uniform behavior in all radial directions, which is in agreement with Figure 4a. As expected, no

significant difference can be observed between the spatial maps for Na^+ , K^+ , Rb^+ , and Cs^+ -containing systems (see Figure S1 in the Supporting Information).

The results presented in this section indicate that the cationic perturbations to the solvent's geometric and electrostatic fluctuations are localized around each cation. Thus, it is unlikely that the experimentally observed change in ET rate is the result of the indirect perturbations from the cation. Also, our similarity order parameter analyses indicate that the cationic perturbation to solvent's electrostatic fluctuations is negligible even for the water molecules immediately adjacent to the cation (Figure 4). We conclude that for monovalent redox centers, specific cation effects for bulk-phase Marcus-like ET rates are not the result of the cation's influence on the aqueous hydrogen network.

3.2. Role of Ion Pairing in Cation-Dependent ET Rates. Specific cation effects can be amplified in the case where redox species pair directly with spectator cations or share a common cation in their solvation shells. The amount of charge on the redox species plays an important role in determining the extent of redox–ion pairing, which can modulate the electrostatic fluctuations in the vicinity of the redox species. To better understand how ion pairing influences these fluctuations, we first quantify the extent of ion pairing as a function of redox valence in Sections 3.2.1 and 3.2.2, and then we look at the effect of ion pairing on the electrostatic fluctuations in Section 3.2.3. For simplicity, we treat the supporting anions (Cl^-) as a generic anionic redox species, and we model higher-valence redox species by artificially increasing its effective charge from -1 to -1.5 , as described in Section 2.2. This allows us to simulate how the variation in redox valence affects the redox–cation pairing behaviors, thereby changing the reaction rate by modifying the electrostatic fluctuations around the redox centers. Analyses on simulations of -2 charged redox centers result in the same conclusion (see the Supporting Information).

3.2.1. Quantifying Ion Pairing Prevalence. Since ion-paired species are expected to move synchronously, we identify ion pairing based on velocity correlations between cation–anion pairs. We define the velocity correlation function as

$$C_{\alpha,\beta} = \frac{1}{N_t} \sum_t \langle \hat{V}_i^{(\alpha)}(t) \times \hat{V}_j^{(\beta)}(t) \rangle \quad (9)$$

where $\hat{V}_i^{(\alpha)}$ is the unit velocity vector for ion i of type α , the angle brackets represent an ensemble average over all the ij pairs at time t , and N_t is the number of timesteps in our time averaging. Using instantaneous velocities in eq 9 tends to generate noisy results as fast ballistic motions could drown out small in a short time but significant in a long time ion pairing behaviors. In order to eliminate these short-time ballistic motions from the analysis, the velocity vector $V_i^{(\alpha)}$ is obtained by considering the displacement of the i th atom of type α , $\Delta x_i^{(\alpha)}$, over a 2 ps period, i.e., $V_i^{(\alpha)} = \Delta x_i^{(\alpha)}/2$. The magnitude of $C_{\alpha,\beta}$, which is a dimensionless quantity due to the fact that we used unit velocity vectors in eq 9, relates to the strength of ion pairing: the larger the value, the stronger the pairing. In our case, α represents the alkali cations and β stands for redox anions. The results for the model redox center with the two effective charges in the presence of different alkali cations are presented in Figure 5. As expected, the redox–cation pairing increases when the valence of the redox center is increased from -1 to -1.5 . In addition, decreasing the charge density of

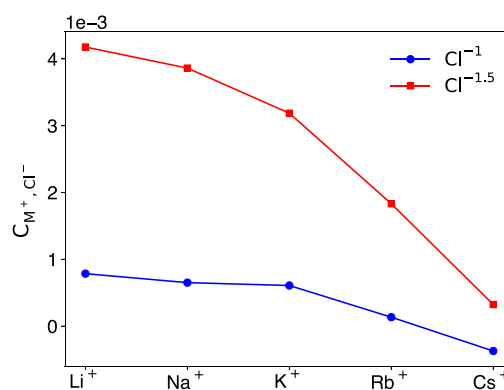


Figure 5. Averaged velocity correlation between the cations and redox centers at different redox charge values.

the cation (from Li^+ to Cs^+) results in a decrease in the degree of redox–cation pairing. This confirms that cation–redox pairing is more pronounced with higher-valence redox centers.

3.2.2. Quantifying Ion Pairing Times. Another quantity that can be used to infer the degree of ion pairing is the residence time, defined as the average amount of time a cation spends in the solvation shell of a given redox center. While the averaged velocity correlation provides a measure of the extent of redox–cation pairing, residence time analysis reveals the duration of ion pairing. We quantify residence time by applying the methodology described in ref 72. Specifically, we utilize an indicator function, $h_j^{(i)}(t_1, t_2)$, that is equal to 1 when molecule j resides in the first solvation shell of the i th redox center at times t_1 and t_2 (with no transient departures longer than 2 ps) and equal to 0 otherwise. We define the average retained coordination number as

$$n(\tau) = \frac{1}{N_{\text{red}} N_t} \sum_i \sum_t \sum_j h_j^{(i)}(t, t + \tau) \quad (10)$$

where the first summation is taken over all N_{red} redox species, the second summation is taken over N_t time periods along an equilibrium trajectory, and the third is taken over all the cations in the solvation shell of the redox centers at the beginning of each time period. From this definition, it follows that $n(0)$ is the standard coordination number, which can also be obtained from integrating the first peak of the RDF.

The time traces of $n(\tau)$ for systems with -1 charged redox centers (redox_{-1}) are plotted in Figure 6a, which shows that the retained coordination numbers of larger cations decay faster with time. All the curves in Figure 6a are normalized to their initial coordination number, $n(0)$, for easier data interpretation. A similar behavior is observed for a higher-valence redox center, as illustrated in Section S4 in the Supporting Information, for systems with -1.5 and -2 charged redox centers. These time traces can be fit using an exponential decay function to obtain the characteristics decay time (or residence time), which is presented in Figure 6b for both the redox_{-1} and $\text{redox}_{-1.5}$ systems. We observe that for each series of systems, the residence time decreases as the cation size increases. As the redox centers have fixed LJ force fields and fixed charges, increasing the cation's radius has the effect of decreasing the equilibrium interaction energy between the cation and redox centers. The smaller attractive forces (negative gradient of the energy) result in the shorter residence times for bigger alkali cations. Following similar reasoning,

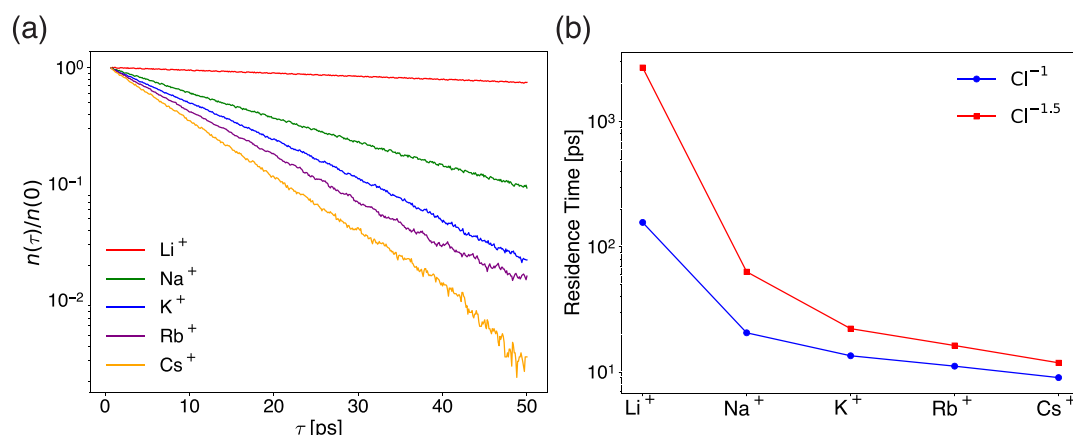


Figure 6. (a) Fraction of cations in the first solvation shell of the redox centers as a function of time for a model redox center with -1 charge. All the curves are normalized using their corresponding standard coordination numbers, $n(0)$. These decay plots are fitted to exponential functions, and the results (residence times) are presented in (b) for a system with -1 and -1.5 charge on the redox centers.

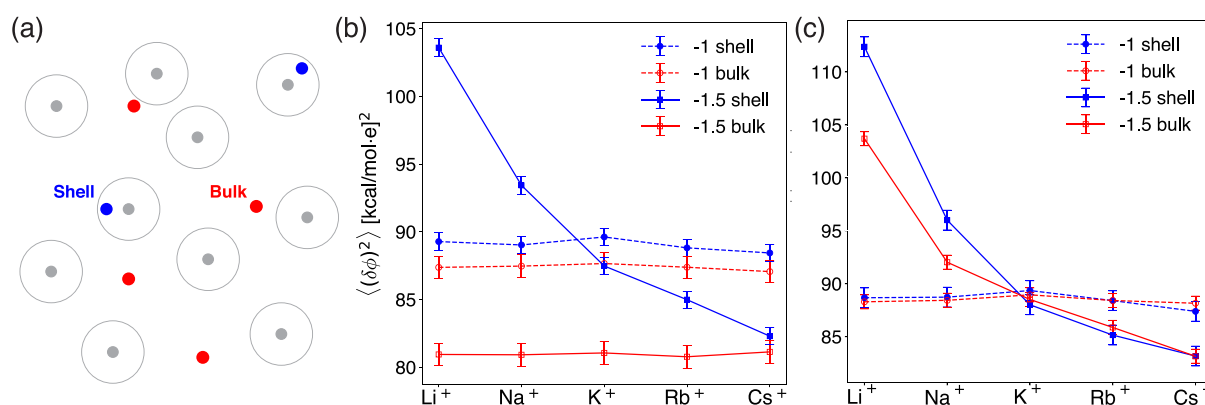


Figure 7. (a) Schematic illustrating bulk vs shell population of redox centers in the simulation. Shell population is defined as the redox centers (blue dots) that are within first two solvation shells of atoms of type α (gray dots). The complementary population of redox centers that are outside the shell population is defined as bulk population (red dots). For this analysis, we choose α to be either cations or anions. For clarity, the cutoff radius of the second solvation shell of each α atom is visualized as gray circles around each gray dot. Following these definitions, panels (b) and (c) show the variances of the Madelung distributions of a general redox center, with -1 [dashed lines] or -1.5 [solid lines] charge, inside [blue lines] and outside [red lines] the second solvation shells of (b) the cations and (c) the anions.

higher redox valence leads to a longer average redox–cation pairing time, which is confirmed by comparing the results for the systems of redox₋₁ and redox_{-1.5} containing the same cations in Figure 6b. The results in Sections 3.2.1 and 3.2.2 show that increasing the redox valence will result in larger extent and longer duration of ion pairing.

3.2.3. Effect of Ion Pairing on the Madelung Profiles of a General Redox Center. From the analyses above, it is clear that higher redox valence increases ion pairing. In this section, we examine how the Madelung potential fluctuations and their sensitivity to different cations change for high-valent redox species. We quantify this change in fluctuations by calculating the variance of Madelung distribution for a model redox center as a function of redox valence.

As we presented in Section 3.1.2, certain solvation effects vary with proximity to cations. Therefore, we separate the redox centers into two subpopulations when studying their Madelung potential distributions, namely, the populations inside and outside the second solvation shells of alkali cations, respectively, for shell and bulk populations, as illustrated in Figure 7a. We define the radial distance cutoff of the second solvation shell as the second minimum of the water–cation radial distribution function. For redox₋₁ systems (blue lines in

Figure 7b), only very small differences can be seen between the Madelung potential variances of two subpopulations, and they appear to be independent from cation identity variation. However, as the valence of redox center increases from -1 to -1.5 while bulk populations retain the same variance across the different cations (red dotted line), the variances of the shell populations show significant dependence on the cation identity, with larger variances in the case of smaller cations (red solid lines). This coincides with the observation of stronger ion pairing between redox centers with smaller cations. Moreover, the fact that the cation dependence emerges for the shell population of redox_{-1.5}, but not redox₋₁, indicates that there is a minimum threshold in the interaction energy for such dependence.

At first glance, the 10 – 20% difference in the Madelung potential variances of the shell population of redox_{-1.5} caused by the different cations seems rather small. However, only the absolute differences (not the ratio) of the variances matter when comparing the trend of a cation-specific rate, as reorganization energy is exponentiated in Marcus rate expression (eq 1). In fact, previous studies have suggested the differences in the reorganization energy in these cation-specific ET reactions to be in the order of 10 kcal/mol,^{23,73}

which is comparable to the absolute differences in variances here ($\sim 20 \times \beta/2 \approx 17$ kcal/mol). To ensure that ion specificity is not diminished by including an extra layer of water molecules (*i.e.*, second solvation shell around the cations), we have performed the same analyses considering only the water molecules in the first solvation shell. The results show very similar trends to the trends in Figures 2 and 7b and are presented in Section S5 in the Supporting Information.

Because electron transfer is most likely to occur when the distance between the two redox centers is small, we also consider the electrostatic fluctuations of redox center pairs that are within each others' first two solvation shells. Here, the shell population refers to those redox centers within the second solvation shell of another redox center, while the bulk population is those farther away from any other redox center. Figure 7c illustrates that for the redox₋₁ case, the shell and the bulk fluctuations are essentially the same, without any dependence on cation identity. However, in the redox_{-1.5} case, the variance of the Madelung potential distribution is cation-dependent for both the shell and the bulk populations. This observation indicates that loosely bound complexes of higher-valence anionic redox pairs and their shared counterions generate local electrostatic fluctuations that are sensitive to the counterion identity. In other words, the reorganization energy becomes sensitive to cation identity when local ion pairing effects are prevalent. Clearly, introducing additional high-valent ions into the solvation shell does not bring a qualitative change to the underlying behavior. Indeed, this charge effect is generalizable for all the higher charged redox centers. The ion pairing and its influence on electrostatic fluctuations become more pronounced when we increase the charge on the redox center to -2 (see Figure S5 Section S4 in the Supporting Information).

4. CONCLUSIONS

In this manuscript, we evaluate the origins of specific cation effects on the rates of simple Marcus-like ET reactions. By considering dilute aqueous electrolyte solutions, we find that the identity of a cation has a negligible influence on the extended water H-bond network, which determines the electrostatic fluctuations on the redox centers that are known to drive ET. Thus, the nominal role of cations as structure makers or structure breakers does not appear to play a role in modifying ET rates. By considering higher-valence redox species, we find that ion pairing likely plays a significant role in both localizing charge in the redox solvation shell and modifying the statistics of local electrostatic fluctuations.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acs.jpcb.0c09895>.

Force field parameters, likelihood function, radial distribution function, tetrahedral order parameter, results (residence time, velocity correlation, and electrostatic fluctuations) for the $-2e$ redox case, and short-range ion-specificity analyses (PDF)

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Notes

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